

JBM9204	INTRODUCTION TO BIOMETRIC SYSTEMS	L	T	P	C
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COURSE OBJECTIVES:

- To familiarize the fundamental principles of different biometric systems
- To emphasize concepts and design of fingerprint recognition biometric systems.
- To know principles of face and hand geometry recognition systems
- To analyze and understand the principles of multimodal biometric system
- To identify issues in realistic evaluation of biometrics - based authentication systems

PREREQUISITE:

- Pattern Recognition and Neural Networks

UNIT I - BIOMETRIC SYSTEMS AND AUTHENTICATION TECHNOLOGIES	9
Biometrics – Authentication vs. Verification, Biometric characteristics - Types of Biometrics – General Architecture of Biometric System and Functionalities – Biometric applications - Authentication technologies – Need for strong authentication – Protecting privacy and biometrics policy –performance measures.	
UNIT II - FINGERPRINT IDENTIFICATION TECHNOLOGY	9
Biometrics using Fingerprint - General description of fingerprints - Fingerprint Recognition system - fingerprint sensors using RF imaging techniques - Finger print feature processing techniques - Minutiae Extraction – Computer enhancement and modelling of fingerprint images - fingerprint classification – fingerprint matching.	
UNIT III - FACE RECOGNITION AND HAND GESTURE RECOGNITION	9
Introduction to face recognition, Algorithms for face recognition - Neural networks for face recognition – Basics of Hand geometry – Image capturing – Hand Segmentation- Feature Extraction - Adaptive Classifiers - Visual-Based Feature Extraction and Pattern Classification.	
UNIT IV - IRIS AND RETINAL BIOMETRICS	9
Iris Image Acquisition using iris sensors. Iris image pre-processing, segmentation, Iris Localization, Daugman's method, iris code, statistical properties of the iris code. Other iris coding methods. Retina scanning: Technical description, Characteristics, Strengths – Weaknesses, Deployment. Retina vascular pattern.	
UNIT V - MULTIMODAL BIOMETRICS AND PERFORMANCE MEASURES	9
Introduction to Multimodal Biometrics –Biometric Fusion- Two Factor Authentication - Normalization- Multimodal Biometrics using Face and Ear – Characteristics and Advantages of Multimodal Biometrics – AADHAAR: An Application of Multimodal Biometrics. Performance evaluation- Statistical Measures of Biometrics – FAR – FRR – FTE – EER.	

TOTAL: 45 PERIODS

COURSE OUTCOMES:

At the end of the course, the student should be able to:

- Demonstrate knowledge on engineering principles underlying biometric systems.
- Extract the finger print features, and apply for finger print recognition.
- Apply different algorithms for face Recognition and hand geometry in biometric application.
- Apply basic knowledge and skills necessary for the Iris and Retinal biometric recognition.
- Elaborate on various multimodal biometrics and performance evaluation.

TEXT BOOKS:

1. Arun A. Ross, Karthik Nandakumar, A.K.Jain, Handbook of Multi biometrics, Springer, New Delhi, 2006. **(Unit 1 to 4)**
2. James Wayman, Anil Jain, Davide Maltoni, Dario Maio, "Biometric Systems, Technology - Design and Performance Evaluation", Springer, 2005. **(Unit 5)**
3. David D. Zhang, Automated Biometrics: Technologies and Systems, Kluwer Academic Publishers, New Delhi, 2000. **(Unit 1 to 5)**

REFERENCES:

1. Anil K. Jain, Ruud Bolle, Sharath Pankanti, "Biometrics, Personal Identification in Network Security", Springer, 2006.
2. D. Maltoni, D. Maio, Anil K. Jain and Salil Prabhakar "Handbook of Fingerprint Recognition", Springer, 2009.
3. Stan Z. Li and Anil K. Jain "Handbook of Face Recognition", Springer; 2nd ed., 2011.
4. John Chirillo, Scott Blaul, "Implementing Biometric Security", John Wiley, 2003.
5. S.Y. Kung, S.H. Lin, M.W.Mak, "Biometric Authentication: A Machine Learning Approach" Prentice Hall, 2005.

WEB REFERENCES:

- <https://sites.google.com/a/di.uniroma1.it/biometric-systems/home-page/course-material>
- https://link.springer.com/chapter/10.1007/0-387-21685-5_2
- <https://machinelearningmastery.com/face-recognition-using-principal-component-analysis/>

Mapping of CO with PO:

	Program Outcomes											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	3	1	2	2	1	3	-
CO2	3	3	3	3	3	3	1	2	2	1	3	-
CO3	3	3	3	3	3	3	1	2	2	1	3	-
CO4	3	3	3	3	3	3	1	2	2	1	3	-
CO5	3	3	3	3	3	3	1	3	3	1	3	-
AVG	3	3	3	3	3	3	1	2.2	2.2	1	3	-